



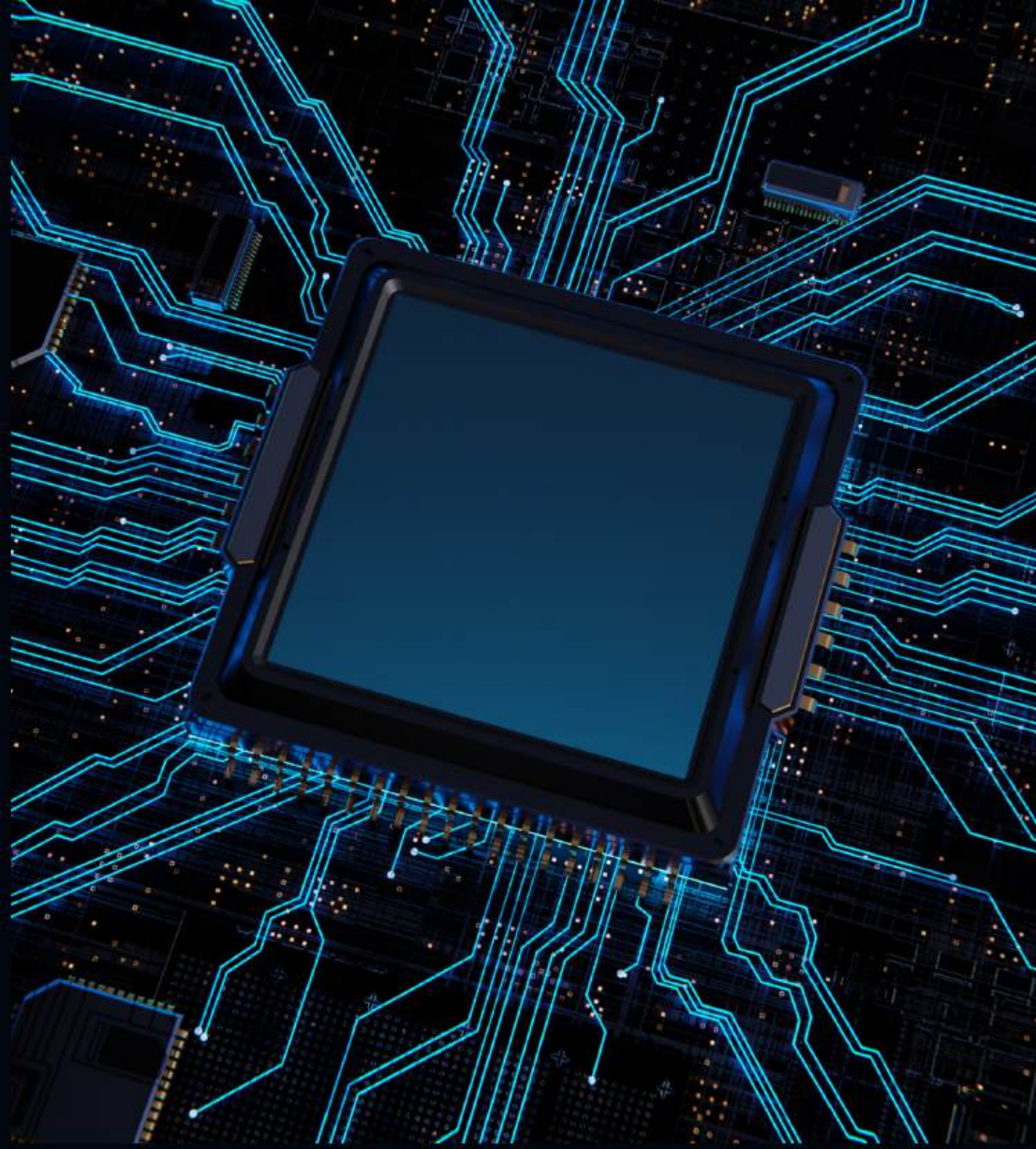
CYBER SECURITY

AI-Powered Cyber Defense System

Real-Time Threat Detection, Response &
Recovery Platform

TEAM NAME :SECUREMIND

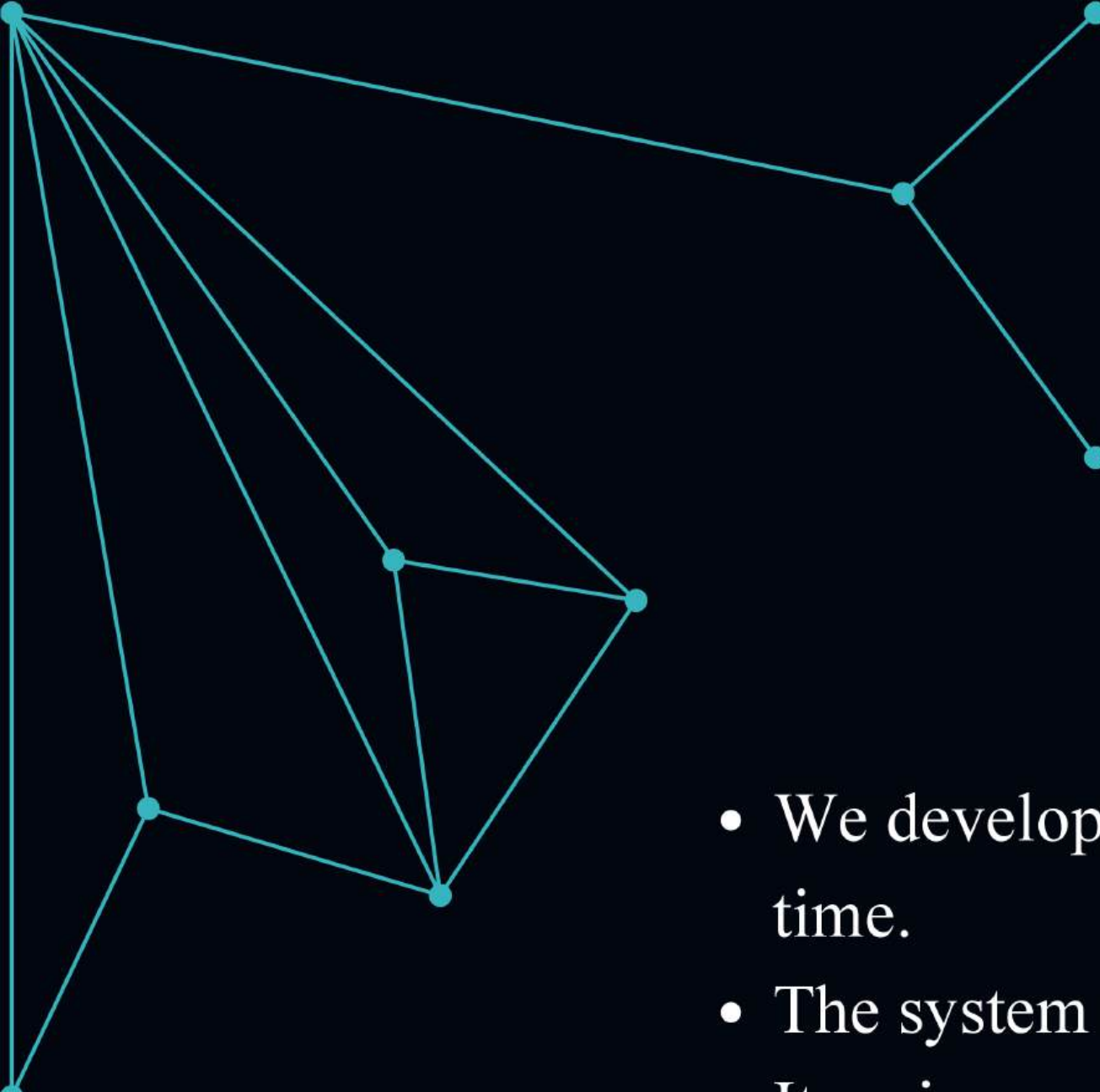
Hackathon: [TECHFUSION 2.0]



PROBLEM STATEMENT

- Cyber attacks such as unauthorized access, fraud, and phishing are increasing rapidly.
- Traditional security systems are not capable of detecting threats in real time.
- Most systems lack user involvement in verifying suspicious activities.
- This results in security breaches, financial loss, and data compromise.





PROPOSED SOLUTION

- We developed an AI-powered cybersecurity system to detect suspicious activities in real time.
- The system uses machine learning to analyze user behavior and identify anomalies.
- It assigns a risk score and triggers alerts when threats are detected.
- A verification layer ensures that only legitimate users can proceed.
- The system also includes automated response and account recovery mechanisms.

SYSTEM ARCHITECTURE



- The system follows a complete cybersecurity pipeline:
- Detection → Risk Scoring → Verification → Response → Recovery
- User activity is continuously monitored and analyzed using a machine learning model.
- Suspicious behavior is identified and assigned a risk level.
- Based on the risk, the system triggers verification and takes appropriate action.

TECHNOLOGY STACK

Frontend: React.js for building interactive user interfaces

Backend: Python Flask for handling APIs and server logic

Machine Learning:

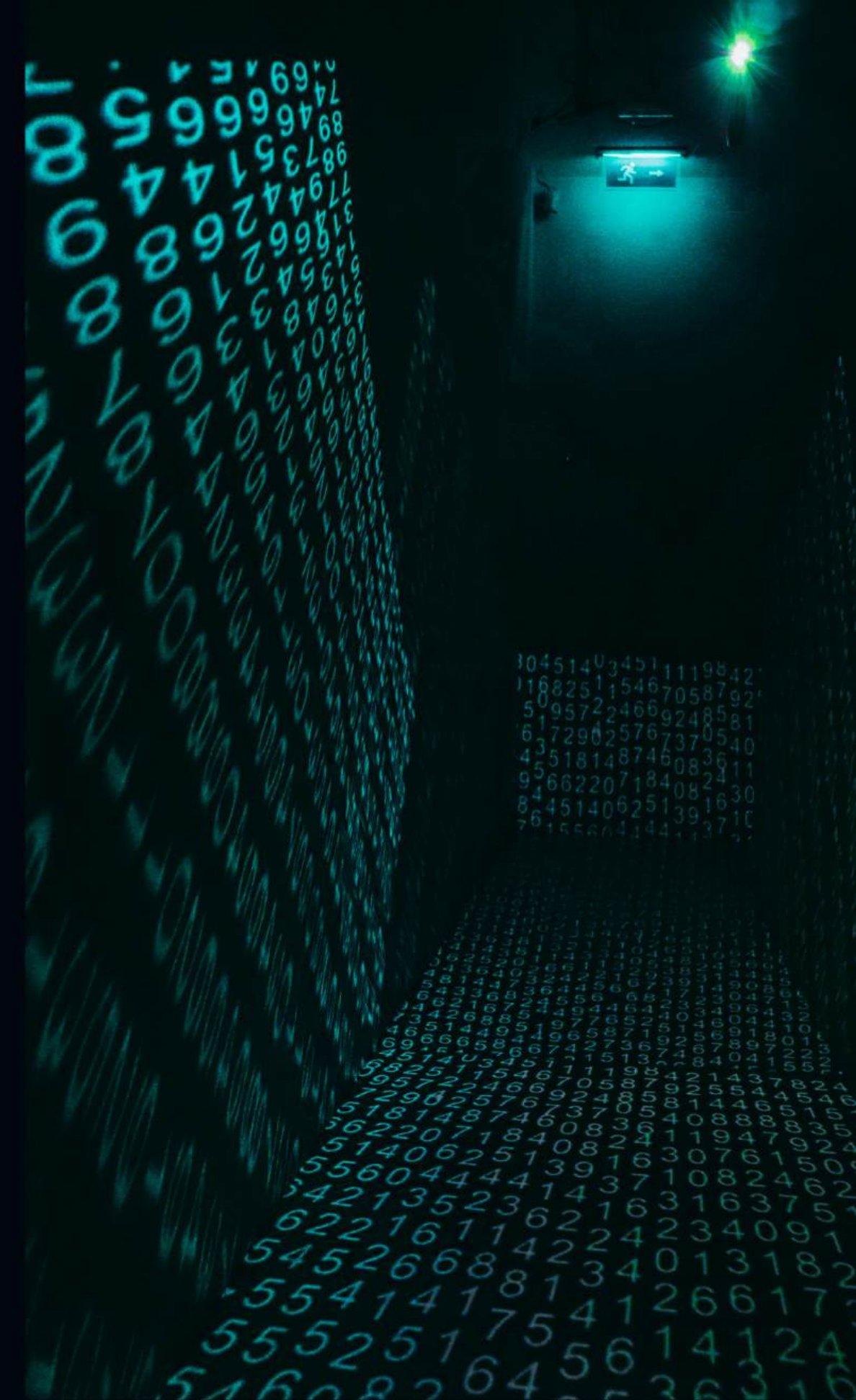
1. LightGBM / Scikit-learn for threat detection
2. Trained using both labeled and unlabeled data
3. Unlabeled network traffic captured using Wireshark

This combination enables accurate and real-time anomaly detection.



Key Features

- Real-time threat detection using machine learning
- Dynamic dashboard displaying live traffic and alerts
- Risk-based classification of attacks (Low / Medium / High)
- User verification using OTP and CAPTCHA (simulated)
- Automated response system to block or allow actions
- Secure account recovery interface for legitimate usersn



CONCLUSION



- We developed an AI-powered cyber defense system for real-time threat detection and response.
- The system integrates detection, verification, automated response, and recovery.
- It improves security by combining AI intelligence with user involvement.
- This solution can be extended for real-world deployment with advanced integrations.



THANK YOU

